

High Power ASE Source Module with Innovative Architecture

Antonin Gallet⁽¹⁾, You Wu⁽¹⁾, Nayla El Dahdah⁽¹⁾, Sheherazade Lamkadmi Azouigui⁽¹⁾ and Romain Brenot⁽¹⁾

⁽¹⁾ Optical Communication Technology Lab, Paris Research Center, Huawei Technologies France, 20 quai du Point du Jour, 92100 Boulogne Billancourt France, antonin.gallet@huawei.com

Abstract: We propose a 50nm and 70mW white source module with innovative design based on a single Semiconductor Optical Amplifier (SOA) with an optical circulator. By combining it with a high power SOA amplifier, we obtain 430mW in-fiber power.

Introduction

Operational margins reduction in wavelength division multiplexing (WDM) systems is key for their performance optimization [1]. To reduce operational margins, the system should be used with all the wavelengths used to design the optical amplifiers. Indeed, the optical amplifier characteristics depends on optical input power and input spectrum shape. In the case where all wavelengths are not deployed, "dummy light" can be employed instead of the real modulated data [2]. Amongst existing approaches, amplified spontaneous emission (ASE) of semiconductor optical amplifier (SOA) can be used to produce dummy light sources. SOA-based ASE source modules have been developed for more than 20 years [3]. However, until recently their performances in term of power consumption and cost were not sufficient for their commercial deployment in WDM systems. Indeed, we have recently proposed a new architecture of ASE source module based on two SOAs in a loop configuration [4]. Such source reaches more than 150 mW power, flat spectrum and low polarization dependence. In order to improve the source efficiency and reduce its cost, we introduce in this paper an innovative source based on the use of a single SOA, a gain flattening filter (GFF), a variable optical attenuator (VOA) and a free-space circulator. All elements are integrated in a packaged module. Such architecture is described in Fig.1 (b) and compared with the traditional scheme (a) that uses 2 SOA as described in [5] and [6]. We will

show that the circulator relaxes the requirement on facet reflectivity to get low gain ripple. Furthermore, we present outstanding optical power of 430 mW by combining the new source with an optical amplifier based on SOA.

ASE source design principle

In this architecture, we use a circulator to inject the emitted ASE back to the SOA. The efficiency of the source is thus improved as the number of SOA is reduced. As illustrated in Fig. 1 (b), the ASE emitted from the left facet is coupled back to the SOA by connecting the SOA to the port 2 of a free space circulator, and connecting ports 1 and 3 of the latter to create a closed loop. In this loop, a Thin-Film Filter GFF and a MEMS-based VOA are inserted to enable spectral flattening and performance tuning via the VOA voltage. The amplified ASE collected at the other SOA facet is split in two using a 50/50 plate. In one arm, the polarization of the light is rotated. Then both polarizations are recombined using a polarization beam combiner (PBC). The output light is hence unpolarized.

Note that the GFF shape is designed to compensate the natural parabolic shape of SOA gain spectrum and to minimize the power outside of targeted spectral range. Regarding the variable optical attenuator, it enables the control of the power level going back to the SOA which in-turn enables the fine-tuning of the module spectrum shape.

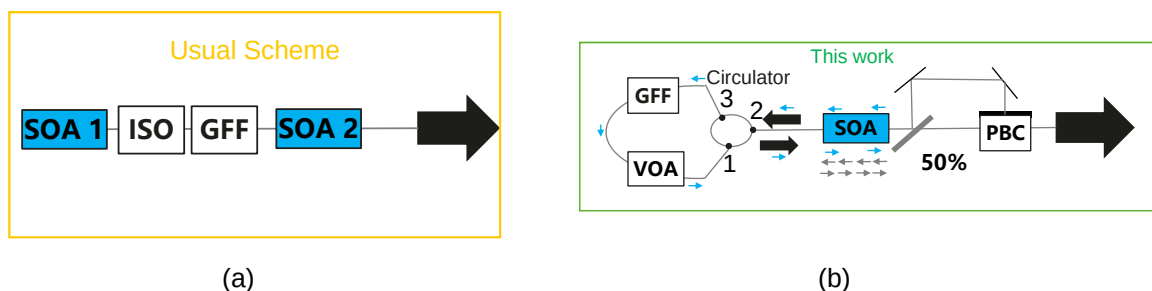


Fig. 1: (a) Traditional SOA based ASE source scheme with two SOAs, implemented in [5] and [6]. An isolator (ISO) and a gain flattening filter (GFF) can be inserted between the two SOA chips. (b) Proposed scheme with circulator, GFF and variable optical attenuator (VOA). After the SOA, an optional polarization diversity scheme is implemented to obtain dual polarization output

SOA Gain and saturation power requirements
in this architecture, the SOA gain should be large in order to have large ASE power. The ASE power should be larger than the input saturation power of the SOA. More precisely, it should be close to the 6dB input saturation power such that the output power is higher than the saturation power P_{sat} . The targeted gain compression is depicted in green the Fig. 2. Going into deep gain

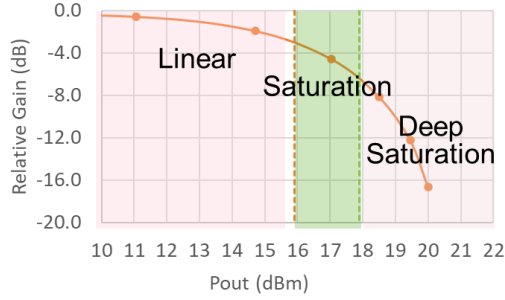
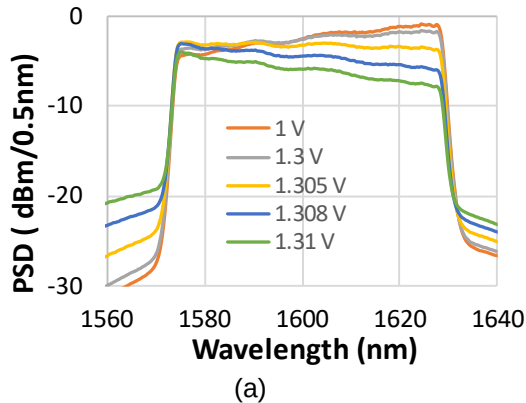


Fig. 2: SOA Saturation requirement

saturation region would reduce the efficiency as a much larger gain would be needed to obtain the required ASE. On the contrary, if the gain of the SOA is too low, the source output power won't reach large power levels. Note that the SOA gain should also compensate the small extra insertion loss of GFF and VOA (1dB for both).

In order to get flat output power, the impact of saturation on power flatness should be assessed. In SOA, Output saturation power usually increase with wavelength. This is mainly due to the wavelength dependance of differential gain. Furthermore, in SOA, the gain broadening is "homogeneous": the gain at every wavelength is reduced when the input power at a specific wavelength increases. The spectrum shape thus depends on the average saturation level. Hence, in the proposed architecture, we use a variable optical attenuator to tune the saturation level so the spectrum shape can be finely adjusted.



SOA Reflectivity requirement

In SOAs, the ripple on the ASE output spectrum is linked to the residual facet reflectivity R_1 and R_2 of the facets. Such cavity is represented with grey arrows on Fig. 1b. As described in [7], we can write for the gain ripple:

$$\Delta G_{dB} = \frac{40}{\ln(10)} \times \sqrt{R_1 R_2} G_0$$

If $R_1 = R_2 = R$, we can express the required reflectivity for a given ripple ΔG_{dB} (usually specified as 0.1 dB) as

$$R = \Delta G_{dB} \times \frac{\ln(10)}{40} \times \frac{1}{G_0}$$

For a 32dB gain SOA, we have $R = 3.6 \times 10^{-6}$. In our module architecture, another cavity is formed by the light going from and to the right facet of the SOA, represented by blue arrows in Fig. 1b.

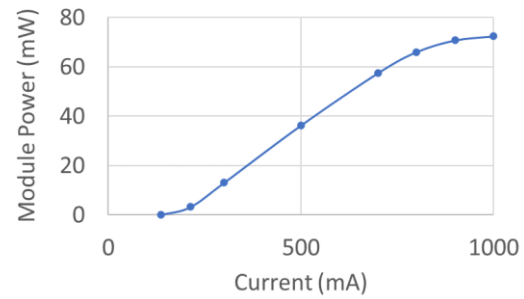


Fig.3: ASE Source module power versus current showing more than 70 mW output power

We thus have

$$R_1 = \Delta G_{dB} \times \frac{\ln(10)}{40} \times \frac{4^2}{G_0^2}$$

Where the factor $4^2 = 16$ comes from the fact that the small signal gain is compressed by 6 dB.

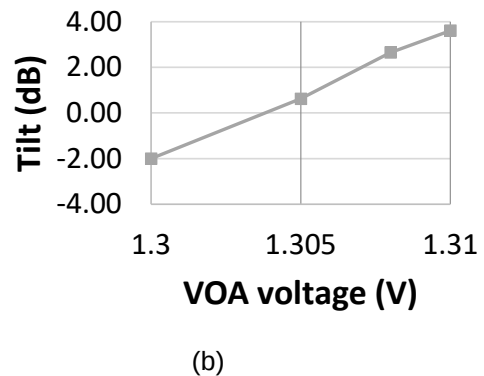


Fig. 4: Effect of VOA on the module spectrum. (a) Spectrum for SOA current of 1A for different VOA voltage. It shows variation mainly on the long wavelength side. (b) Extracted tilt showing variations from -2 to 4 dB.

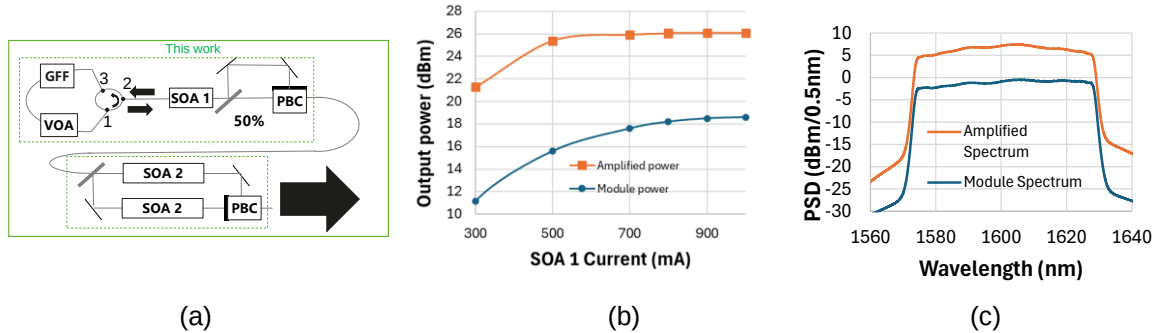


Fig. 5: ASE Source amplified by SOA module. (a) Schematic of the experiment (b) Output power as a function of SOA1 current with SOA module (orange curve) and without (blue curve). SOA2 are biased at 1.5A. (c) Spectrum for a SOA1 current of 1A with the SOA2 biased at 1.5A (orange curve) and without the SOA module (blue curve)

We obtain $R_1 = 2.5 \times 10^{-7}$. As described in [8], we can reach this value using a 7° tilted SOA with standard anti-reflection coating. One can note that if we had used a 100% reflective mirror instead of the circulator as described in [9], the required reflectivity would have been lower (by 3 orders of magnitude) which would be challenging to achieve.

SOA Fabrication

The SOA is made using semi-insulating buried heterostructure with asymmetric cladding structure on Indium Phosphide (InP). SOA fabrication begins with the epitaxial growth of an n-doped InGaAsP slab, the active region (three 5 nm thick InGaAsP Quantum Well) and the p-doped top layer. The layers are etched to form a waveguide. Then semi-insulating and p-doped regrowth are made. After bar cleaving and chip separation, individual SOAs are mounted on Aluminum Nitride carrier.

ASE Source results

The ASE source was integrated in a module including a Thermo-electric cooler for temperature control. The ASE source was operated at 45°C temperature. With SOA bias current increase, both gain and saturation power increase. Thus, the module output power also increases. At 1A, the gain is 32 dB and 3 dB output saturation power is 15.5 dBm at $1.6 \mu\text{m}$. As depicted in fig. 3, we note a linear increase of the output power up to more than 70 mW (18.6 dBm). The ASE power level going back to the SOA is tuned thanks to the VOA placed between the port 2 and port 3 of the circulator. The impact of VOA on spectrum is presented on Fig. 4. For this figure, the SOA is biased at 1A and the VOA voltage is swept between 1 and 1.31V. The tilt, defined as the power at long wavelength minus the power at short wavelength gradually move from -2 to 4 dB. As seen on the Fig. 4 (a) and Fig.4 (b), the output power is flat (less than 0.5 dB variation over the

spectral band) for 1.305V VOA voltage. When the VOA voltage increases, the power at the short wavelength side increases by up to 1dB. This is due to the global saturation decrease of the SOA due to VOA attenuation.

ASE Source amplified with SOA module

To obtain even higher output power, we have connected the output of our module to a dual polarization high power SOA module developed for in-line amplification [10] shown in Fig. 5 (a). The SOA module has 16 dB gain at $1.6 \mu\text{m}$ and a 3 dB output saturation power above 24 dBm, for 1500 mA drive current in each SOA. The module is operating at 25°C . The power and spectrum of such module fed with the ASE source is shown in Fig. 5 (b) and 5 (c). The maximum power is more than 26 dBm and the amplified spectrum is flat. Such power is nearly 3 times higher than reported in [4]. As we can see on Fig. 5(b), the power has nearly reach its maximum value for a ASE source SOA of 700 mA. Indeed, the gain has already reach its 6 dB saturation point for this current. The emitted power nearly reaches the 450 mW reported in [5] with 3 SOAs instead of 6, demonstrating the interest of this architecture.

Conclusions

We have demonstrated an innovative ASE source module based on a single SOA and circulator with more than 70 mW output power and spectral tilt capability. When combined with a high power SOA module, up to 430 mW power with flat output is obtained. In the future, we may develop new SOA structures to evaluate such configuration for forward Raman pump application.

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